

EGRA and EGMA Subtasks Included in AHEAD

1. Introduction

The following tables provide descriptions and key information for the core and common subtasks of the **Early Grade Reading Assessment (EGRA)** and the **Early Grade Mathematics Assessment (EGMA)**, as well as the additional subtasks included in the assessment surveys contained in the AHEAD harmonised dataset.

Although EGRA and EGMA are adapted to the language, curriculum, and educational context of each country, a common set of subtasks has become standard across most implementations. Many assessment programmes also include supplementary subtasks to measure additional literacy or numeracy skills or to address specific programme, curriculum, or research objectives. Accordingly, the AHEAD harmonised dataset includes both these widely implemented core subtasks and the additional subtasks administered in individual assessment studies.

The subtask descriptions have been compiled from available documentation and standardised to facilitate comparison across projects. Every effort has been made to ensure that the descriptions accurately reflect the original assessment instruments and administration procedures. However, they are based on the documentation currently available and may be revised as additional project documentation, assessment instruments, or technical manuals become available.

2. Core and common EGRA Subtasks

The Early Grade Reading Assessment (EGRA) is an individually administered assessment designed to measure the foundational reading skills that children develop during the early years of primary school. EGRA consists of a series of short subtasks that assess different components of reading development. Although the number and content of subtasks vary across languages and assessment programmes, they generally progress from foundational skills, such as letter-sound knowledge and word recognition, to more advanced skills, including oral reading fluency and reading comprehension.

The table below describes the core and common EGRA subtasks included in the AHEAD harmonised dataset. The variable name column presents the harmonised variable names used in the AHEAD dataset; individual assessment studies may use different variable names in their original datasets. Similarly, the timing limits reported for each subtask reflect the most common EGRA implementations. Individual studies may adopt different time limits, discontinuation rules, or administration procedures depending on the language, curriculum, and assessment design.

Table 1: Core and common EGRA subtasks

Task name	Skill	Description	Variable name
Letter name identification	Alphabetic principle - decoding	Learners are presented with a randomised series of upper- and/or lower-case letters and asked to orally identify each by name. Performance is scored for both accuracy and rate and reported as letter names correctly identified within 60 seconds (letter_name_pm).	letter_name
Letter sound identification	Alphabetic principle - decoding	Learners are presented with a randomised series of upper- and/or lower-case letters and asked to orally identify each by its corresponding sound. Performance is scored for both accuracy and rate and reported as letter sounds correctly identified within 60 seconds (letter_sound_pm).	letter_sound
Invented word reading	Alphabetic principle - decoding	Learners are presented with a written list of invented words that follow the phonological and orthographic rules of the language but are not real words. Learners read aloud as many as possible within 60 seconds. Performance is scored for both accuracy and rate and reported as correct invented words per minute (invent_word_pm).	invent_word
Oral reading fluency	Reading fluency	Learners read a grade-appropriate passage of approximately 50–60 words aloud as accurately and quickly as possible within 60 seconds. Performance is scored for both accuracy and rate and reported as correct words read per minute (oral_read_pm).	oral_read
Listening comprehension	Listening comprehension / Oral language	Learners listen to a short passage read aloud by the assessor and respond to a series of explicit and inferential comprehension questions. Responses are scored for accuracy.	list_comp

Familiar word reading	Word recognition	Learners are presented with a written list of 50 high-frequency, grade-appropriate words in isolation and asked to read aloud as many as possible within 60 seconds. Performance is scored for both accuracy and rate and reported as correct familiar words read per minute (fam_word_pm).	fam_word
Reading comprehension	Reading comprehension	Learners read a grade-appropriate passage silently or aloud and respond to comprehension questions. Responses are scored for accuracy.	read_comp
Vocabulary	Oral vocabulary	Learners are orally presented with a series of words and asked to identify their meanings, typically by pointing to the corresponding object, picture, or body part. Responses are scored for accuracy based on the number of correct vocabulary words identified.	vocab_oral
Initial sound identification	Phoneme awareness	Learners are orally presented with a word and asked to isolate and pronounce its first sound. Responses are scored for accuracy based on correct phoneme identification.	pa_init_sound
Different sound identification	Phonemic awareness	Learners listen to a series of spoken sounds, syllables, or words and identify the sound that is different from the others. Responses are scored for accuracy.	pa_di_sound

3. Typical EGRA Administration Procedures

Although EGRA implementations vary across countries and languages, the core subtasks are generally administered individually by a trained assessor using a standardised administration protocol. Assessors follow scripted instructions, use standardised stimulus materials, and record learner responses according to predefined scoring rules. Although the administration procedures described below are common across many EGRA implementations, individual studies may differ in their timing limits, discontinuation rules, scoring procedures, and administration protocols.

Examples of the stimulus materials used for selected EGRA subtasks are provided below for illustrative purposes. These examples represent common task formats rather than the exact materials used in every assessment. Additional examples are available in the EGRA Toolkit

(RTI International, 2016) and the where available in the original assessment documentation for individual studies included in the AHEAD harmonised dataset.

Letter name and sound identification

These subtasks are typically discontinued if the learner is unable to correctly identify any of the first ten letters (i.e., the first row of letters), although discontinuation rules may vary slightly across assessment programmes. The EGRA administrator times the learner for 60 seconds, recording the number of correctly identified letters as well as any errors made while the learner responds aloud.

Figure 1: Letter-sound identification subtask (Wolof language, Senegal)

Misaal :	o	uu	t	mb					
a	n	o	m	i	u	d	e	L	k
g	b	y	a	uu	x	t	w	n	f
s	r	c	j	p	ñ	oo	i	à	m
n	k	aa	r	ã	ee	ée	d	u	r
ee	ii	b	L	g	y	oo	L	a	t
óó	k	u	a	m	x	é	t	n	c
i	g	ë	j	d	a	e	aa	ng	o
y	x	a	L	m	nd	t	s	nj	L
η	w	u	b	ë	n	i	a	y	e
i	a	aa	k	d	U	o	mb	e	i
Misaal	di		bu		dem				

Source: RTI, 2016

Figure 2: Letter-sound identification task EGRS South Africa

B	m	S	f
---	---	---	---

m	l	h	g	S	y	Z	W	p	e
L	k	T	D	b	a	c	o	n	ng
i	b	th	M	U	r	j	dl	K	u
g	R	B	kh	l	f	hl	M	s	kw
S	N	ph	B	p	v	k	a	E	D
R	A	t	P	f	sh	h	u	a	t
dw	G	H	B	S	l	g	m	i	j
B	dl	o	m	O	Y	E	N	p	t
g	K	B	ny	y	bh	Z	V	D	nc
f	s	ng	a	z	p	C	th	G	sw
V	ncw	q	h	nhl	g	sh	y	kh	t

Source: Department of Basic Education and the University of the Witwatersrand (2018)

Familiar and invented word reading

These subtasks are typically discontinued if the learner is unable to correctly read any of the first five words, although individual studies may use alternative discontinuation rules. The EGRA administrator times the learner for 60 seconds, recording the number of correctly read words and any reading errors. Examples of how these subtasks may be presented are shown below.

Figure 3: Invented word reading (Icibemba language, Zambia)

- ✎ (/) Mark any incorrect words with a slash
 (Ø) Circle self-corrections if you already marked the word incorrect
 (]) Mark the final word read with a bracket

Examples: opa toti maba

1	2	3	4	5	
lebi	ndite	luti	oya	lusi	(10)
mibu	kibe	shuti	tobe	njolo	(20)
angi	shipe	nomi	sani	opu	(30)
nepa	wipi	tupu	naye	koi	(40)
tate	shuma	telu	shingu	yoba	(50)
seni	nupa	etu	kika	onu	(60)
sale	pafu	tawe	ebi	ewa	(70)
ipa	ombi	kendi	ngopa	ndika	(80)
afu	yema	mawe	tebi	folo	(90)
fimu	yapo	tibu	bife	lefu	(100)

✎ Time remaining on stopwatch at completion (number of SECONDS)

✎ Exercise discontinued because the child had no correct answers in the first line

Source: RTI, 2016

Figure 4: Familiar word reading task EGRS South Africa

sat	dad
-----	-----

dog	rat	ball	run	can	box
big	day	book	tea	yes	seed
ring	lock	cats	hen	clap	fish
and	went	best	from	but	look
find	play	into	under	tell	good
think	drum	black	jumped	singing	green
water	first	three	must	sheep	sleeping

Source: Department of Basic Education and the University of the Witwatersrand (2018)

Oral reading fluency and reading comprehension

The oral reading fluency subtask is typically discontinued if the learner is unable to correctly read any of the first line of the passage (generally around ten words). Most EGRA implementations administer this subtask for 60 seconds, although a small number of studies use alternative timing limits (e.g., three minutes). During administration, the assessor records the number of correctly read words and any reading errors.

The reading comprehension subtask is usually administered only if the learner attempts the oral reading passage. Depending on the assessment protocol, learners may or may not be permitted to refer back to the passage while answering the comprehension questions. Responses are scored for accuracy.

Figure 5: Oral reading fluency and reading comprehension subtask EGRS South Africa

Jabu and his dog	
Jabu had a little dog. It was a happy dog.	10
Jabu liked his dog.	14
One day Jabu and his dog walked in the bush behind the house.	27
The dog saw a rabbit and chased it.	35
The dog got lost.	39
Jabu called him but he didn't come.	46
Jabu was sad. He went home.	52
But before evening the little dog came back.	60
Jabu was very happy. He gave his dog a bone.	70

Task 6.2. English text reading comprehension

Question	Word	Answer
1. 1. What is the name of the boy in the story?	1	Jabu
2. 2. What kind of animal did he have?	5	A dog.
3. 3. What did Jabu and his dog do together one day?	27	They went for a walk/they walked in the bush/They walked behind the house.
4. 4. What animal did the dog chase?	35	A rabbit.

Source: Department of Basic Education and the University of the Witwatersrand (2018)

Listening comprehension

Listening comprehension is generally administered independently of the oral reading tasks. The assessor reads a short passage aloud while the learner listens and then responds to a series of comprehension questions. The task is typically untimed, is not subject to a discontinuation rule, and responses are scored for accuracy.

Figure 6: Listening comprehension (English)

Sub-test 5. LISTENING COMPREHENSION		X		⌚ X
🔊 Merebekenkan abasem tiawa bi baako pe dennennen akyere wo, na mabisabisa wo nsem kakra afa ho. Mesre wo tie no yiye na bua nsemmisa no senea wubetumi biara. Wubetumi de kasa biara a wope ayiyi nsemmisa no ano. Metumi afi ase? Yemfi ase. I am going to read you a short story aloud ONCE and then ask you some questions. Please listen carefully and answer the questions as best as you can. You can answer the questions in whichever language you prefer. Ready? Let's begin.				Remove the pupil stimuli booklet from the child's view. Do not allow the child to look at the passage or the questions. If a child says "I don't know," mark as incorrect.
👉 (✓) 1 = Correct (✓) 0 = Incorrect (✓) . = No response.				
Issa was very sad. He lost his grandfather's sheep. He could not go to look for them. Grandfather came to look for them. Soon he returned with the sheep. Issa is smiling now.				
Why was Issa sad? [he lost his sheep; he could not go to look for his sheep]	1	0	.	
Who went to look for the sheep? [Grandfather]	1	0	.	
Why is Issa smiling now? [Grandfather returned with his sheep; his sheep are back; Grandfather found the sheep]	1	0	.	
Mo! Woaye ade. Yenko ofa a edi so no so. Good effort! Let's go on to the next section.				

Source: RTI, 2016

Phoneme awareness

Phonemic awareness tasks vary across EGRA implementations but generally assess learners' ability to identify, isolate, or manipulate individual sounds in spoken words. Common subtasks include initial sound identification and initial sound discrimination, while some studies also include phoneme or syllable segmentation tasks. These subtasks are typically administered orally, are untimed, and responses are scored for accuracy.

Figure 7: Phonemic awareness – Initial sound identification (English)

SUBTASK 2. PHONEMIC AWARENESS					📖 X	🕒 Untimed																																																																		
👂 This is a listening exercise. I want you to tell me the first sound of each word. For example, in the word “pot”, the first sound is /p/. I would like you to tell me the first sound you hear in each word. I will say each word <u>two times</u>. Listen to the word, then tell me the very first sound in that word. Let’s practice. What is the first sound in “mouse”? ... “mouse”? [If the child responds correctly, say:] Very good, the first sound in “mouse” is /m/. [If the child does not respond correctly, say:] Listen again: “mouse”. The first sound in “mouse” is /mmm/. Now let’s try another one: What is the first sound in “day”? ... “day”? [If the child responds correctly, say:] Very good, the first sound in “day” is /d/. [If the child does not respond correctly, say:] Listen again: “day”. The first sound in “day” is /d/. Ready? Let’s begin.					Read the instructions to the child and conduct the examples. Read the prompt and then pronounce the word a second time. Pronounce each word slowly. 🔄 If the child does not respond after 3 seconds, mark as “No response” and say the next prompt. 👉 If the child responds incorrectly or does not respond to the first five words, say “Thank you!, discontinue this subtask, check the box at the bottom of the page, and go on to the next subtask.																																																																			
<table border="1"> <thead> <tr> <th></th> <th>Item</th> <th>Answer</th> <th>Correct</th> <th>Incorrect</th> <th>No response</th> </tr> </thead> <tbody> <tr> <td>1.</td> <td>What is the first sound in “at”? ... “at”?</td> <td>/a/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>2.</td> <td>What is the first sound in “so”? ... “so”?</td> <td>/s/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>3.</td> <td>What is the first sound in “chalk”? ... “chalk”?</td> <td>/ch/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>4.</td> <td>What is the first sound in “very”? ... “very”?</td> <td>/v/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>5.</td> <td>What is the first sound in “car”? ... “car”?</td> <td>/k/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>6.</td> <td>What is the first sound in “for”? ... “for”?</td> <td>/f/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>7.</td> <td>What is the first sound in “man”? ... “man”?</td> <td>/m/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>8.</td> <td>What is the first sound in “ox”? ... “ox”?</td> <td>/o/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>9.</td> <td>What is the first sound in “yes”? ... “yes”?</td> <td>/y/</td> <td></td> <td></td> <td></td> </tr> <tr> <td>10.</td> <td>What is the first sound in “go”? ... “go”?</td> <td>/g/</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>								Item	Answer	Correct	Incorrect	No response	1.	What is the first sound in “at”? ... “at”?	/a/				2.	What is the first sound in “so”? ... “so”?	/s/				3.	What is the first sound in “chalk”? ... “chalk”?	/ch/				4.	What is the first sound in “very”? ... “very”?	/v/				5.	What is the first sound in “car”? ... “car”?	/k/				6.	What is the first sound in “for”? ... “for”?	/f/				7.	What is the first sound in “man”? ... “man”?	/m/				8.	What is the first sound in “ox”? ... “ox”?	/o/				9.	What is the first sound in “yes”? ... “yes”?	/y/				10.	What is the first sound in “go”? ... “go”?	/g/			
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Source: RTI, 2016

4. Core EGMA Tasks

The Early Grade Mathematics Assessment (EGMA) is an individually administered assessment designed to measure the foundational numeracy skills that children develop during the early years of primary school. EGMA consists of a series of short subtasks that assess different components of mathematical development, including number recognition, numerical magnitude, arithmetic, mathematical patterns, geometry, and problem solving.

As with EGRA, the specific subtasks vary across countries and assessment programmes to reflect local curricula, languages of instruction, and research objectives, although a common set of core subtasks has been adopted across many implementations.

The table below describes the core EGMA subtasks included in the AHEAD harmonised dataset. The variable name column presents the harmonised variable names used in the AHEAD dataset; individual assessment studies may use different variable names in their original datasets. Similarly, the timing limits reported for each subtask reflect the most common EGMA implementations. Individual studies may adopt different time limits, discontinuation rules, scoring procedures, or administration protocols depending on the assessment design.

Task name	Skill	Description	Variable name
Number identification	Number competence	Learners are presented with a series of numerals and asked to identify each number aloud.	num_id
Number discrimination	Numerical magnitude	Learners are presented with pairs of numbers and asked to identify the greater quantity.	quant_comp
Missing number	Mathematical patterns	Learners are presented with numerical sequences containing one missing value and asked to identify the missing number.	miss_num
Addition level 1	Arithmetic	Learners solve simple written addition problems of increasing difficulty within 60 seconds. Performance is scored for accuracy.	addlvl1
Subtraction level 1	Arithmetic	Learners solve simple written subtraction problems of increasing difficulty.	sublvl1
Addition level 2	Arithmetic	Learners solve more complex written addition problems.	addlvl2
Subtraction level 2	Arithmetic	Learners solve more complex written subtraction problems.	sublvl2
Word problems	Arithmetic and mathematical reasoning	Learners listen to or read short mathematics problems presented in everyday contexts and solve each problem.	word_problem

5. Typical EGMA Administration Procedures

Although EGMA implementations vary across countries and assessment programmes, the core subtasks are generally administered individually by a trained assessor using a standardised administration protocol. Assessors follow scripted instructions, present standardised stimulus materials, and record learner responses according to predefined scoring procedures. While the administration procedures described below are common across many EGMA implementations, individual studies may differ in their discontinuation rules, administration procedures, scoring methods, and progression between subtasks.

Examples of the stimulus materials used for selected EGMA subtasks are provided below for illustrative purposes. These examples represent common task formats rather than the exact materials used in every assessment. Additional examples are available in the EGRA Toolkit (Platas et al, 2014) and, where available, in the original assessment documentation for individual studies included in the AHEAD harmonised dataset.

Number identification and number discrimination

The tasks are typically untimed and are discontinued after four consecutive incorrect responses. Responses are scored for accuracy.

Figure 8: Number identification subtask

Number Identification items				
2	9	0	12	30
22	45	39	23	48
91	33	74	87	65
108	245	587	731	989

Source: Platas et al, 2014

Figure 9: Number discrimination subtask

Number Discrimination items			
7	5	94	78
11	24	146	153
39	23	287	534
58	49	623	632
65	67	867	965

Source: Platas et al, 2014

Missing number and Addition and Subtraction Level 1

In a standard EGMA administrations, the missing number subtask is timed for 60 seconds, although adaptations may use different timing limits. Assessors record the accuracy of learners' responses.

The addition level 1 and subtraction level 1 subtasks are each administered for 60 seconds, although adaptations may use different timing limits. Assessors record both the accuracy of learners' responses and, where applicable, the problem-solving strategy used (e.g., mental calculation, fingers or tick marks, or paper and pencil). The subtraction level 1 subtask is generally administered only to learners who successfully complete the preceding addition level 1 subtask, with learners scoring zero on addition level 1 typically not progressing to this task.

Figure 10: Simple Addition and subtraction subtask

$1 + 3 = \square$	$4 - 1 = \square$
$3 + 2 = \square$	$5 - 2 = \square$
$6 + 2 = \square$	$8 - 2 = \square$
$4 + 5 = \square$	$9 - 5 = \square$
$3 + 3 = \square$	$6 - 3 = \square$

Source: Platas et al, 2014

Figure 11: Missing number subtask

Missing Number items

5	6	7	
14	15		17
20		40	50
	300	400	500
2	4	6	

Source: *Platas et al, 2014*

Addition Level 2 and Subtraction Level 2

These subtasks assess more advanced arithmetic skills involving multi-digit calculations. In standard EGMA administrations, each task is administered for 60 seconds, although adaptations may use different timing limits. Assessors record both the accuracy of learners' responses and, where applicable, the problem-solving strategy used (e.g., mental calculation, fingers or tick marks, or paper and pencil). The addition level 2 subtask is generally administered only to learners who successfully complete the preceding subtraction level 1 subtask.

Figure 12: Addition and subtraction level 2 subtask

Addition and subtraction level 2 items	
$13 + 6 = \square$	$19 - 6 = \square$
$18 + 7 = \square$	$25 - 7 = \square$
$12 + 14 = \square$	$26 - 14 = \square$
$22 + 37 = \square$	$59 - 37 = \square$
$38 + 26 = \square$	$64 - 26 = \square$

Source: *Platas et al, 2014*

Word problems

The assessor reads each problem aloud while the learner listens and provides an oral response. Learners may use mental calculation, fingers or tick marks, or paper and pencil to solve the problems, and assessors record both the accuracy of responses and, where applicable, the problem-solving strategy used. The task is typically untimed and responses are scored for accuracy.

Figure 13: Word problems subtask

Problem Type	Example
Change: Result Unknown	Two children are on the bus, three more children get on. How many children are on the bus altogether?
Combine: Result Unknown	There are six children on the bus, two are boys. The rest are girls. How many girls are there on the bus?
Compare: Change Unknown	There are two children on John's bus and seven children on Mary's bus. How many children must join John's bus so that it has the same number of children as Mary's bus?
Change: Start Unknown	Five children get on the bus. Now there are 12 children on the bus. How many children were on the bus to begin with.
Sharing	Four children share twelve candies equally between themselves. How many candies does each child get?
Multiplicative	There are five seats on the bus. There are two children on each seat. How many children are on the bus altogether?

Source: Platas et al, 2014

6. Additional (non-core) EGRA and EGMA subtasks

Additional (non-core) subtasks are included in some EGRA and EGMA implementations to assess literacy and numeracy skills beyond the standard core assessment framework or to address specific programme, curriculum, or research objectives. These subtasks vary across studies in their content, administration, and scoring and are therefore not administered consistently across all assessments. The descriptions below summarise the common characteristics of each supplementary subtask across the assessment studies included in the AHEAD harmonised dataset. The variable name column presents the harmonised variable names used in the AHEAD dataset; individual assessment studies may use different variable names in their original datasets.

Table 3: Additional (non-core) EGRA subtasks

Task name	Skill	Description	Variable name
Differential initial sound	Phonemic awareness	Learners are orally presented with a set of words and asked to identify the word that begins with a different initial sound. Responses are scored for accuracy based on correct identification of the differing initial phoneme.	pa_df_init_snd

Receptive listening comprehension	Listening comprehension	Learners listen to words, sentences, or short passages presented orally by the assessor and demonstrate comprehension by selecting, pointing to, or indicating the correct response.	receptive_listen
Written comprehension	Written comprehension	Learners read written text independently and respond to questions or complete tasks that assess their understanding of the text. Responses are scored for accuracy.	written_comp
Phoneme combined	Phonemic awareness	Learners complete one or more phonemic awareness tasks, including identifying, segmenting, blending, adding, deleting, or substituting individual phonemes in spoken words without reference to print. Responses are scored for accuracy.	phoneme_combined
Creative writing	Written composition	Learners produce original written text in response to a prompt or stimulus. Responses are assessed using predefined scoring criteria that may include organisation and development of ideas, vocabulary, language use, spelling, grammar, and writing conventions.	creative_writing
Untimed oral reading	Connected text reading	Learners read a grade-appropriate passage aloud as accurately as possible without a time limit. Performance is scored for reading accuracy, and in some assessment's additional	unt_oral_read

		measures such as errors or total words read may also be recorded.	
Word dictation	Writing	Learners listen to individual words read aloud by the assessor and write each word.	dict_word
Sentence dictation	Writing	Learners listen to sentences read aloud by the assessor and write each sentence.	dict_sent
Untimed reading comprehension	Reading comprehension	Learners read a grade-appropriate passage and respond to comprehension questions without a time limit. Responses are scored for accuracy. Learners may or may not be permitted to refer back to the passage depending on the assessment protocol.	unt_read_comp
Silly sentences	Sentence comprehension	Learners read a series of simple sentences and determine whether each sentence is meaningful or nonsensical. Responses are scored for accuracy based on correct identification of meaningful and nonsensical sentences.	silly_sentence

Table 4: Additional (non-core) EGMA subtasks

Task name	Skill	Description	Variable name
Written addition	Arithmetic	Learners solve written addition problems of increasing difficulty. Responses are scored for accuracy based on the number of correctly solved items.	we_add

Written subtraction	Arithmetic	Learners solve written subtraction problems of increasing difficulty. Responses are scored for accuracy based on the number of correctly solved items.	we_sub
Written multiplication	Arithmetic	Learners solve written multiplication problems involving single- and/or multi-digit whole numbers. Responses are scored for accuracy based on the number of correctly solved items.	we_mult
Written division	Arithmetic	Learners solve written division problems involving whole numbers. Responses are scored for accuracy based on the number of correctly solved items.	we_div
Multiplication	Arithmetic	Learners solve multiplication problems involving single- and/or multi-digit whole numbers. Responses are scored for accuracy based on the number of correctly solved items.	mult
Fractions	Arithmetic	Learners identify, compare, represent, and solve problems involving fractions. Responses are scored for accuracy based on the number of correctly solved items.	frac
Shape identification	Geometry	Learners identify and name common two- and/or three-dimensional geometric shapes. Responses are scored for accuracy based on the number of correctly identified shapes.	shape_id

7. EGRA and EGMA subtask naming conventions

Every EGRA/EGMA subtask shares a **subtask stem**; individual fields add an item number or metric suffix. Within each **subtask stem**, you will typically find:

Table 4: EGRA/ EGMA subtask naming conventions

Field type	Pattern	Examples
Item-level	{subtask_stem}_{n}	letter_sound_1, num_id_3
Subtask summary	{subtask_stem}_{metric_suffix}	oral_read_score, oral_read_pm, fam_word_time_remain

The full set of metric suffixes is shown in the table below.

Table 5: Standard metric suffixes

Metric suffix	Description	Timed tasks	Untimed tasks

_score	Raw score	all	all
_pm	Correct items per minute	all	—
_score_pcmt	Percent correct	—	all
_attempted	Items attempted	all	reading comprehension
_attempted_pcmt	Percent correct of items attempted	all	reading comprehension
_score_zero	Indicator: score equals zero	all	all
_time_remain	Time remaining (seconds)	all	—
_auto_stop	Auto-stop rule triggered	all	—
_max	Maximum items in subtask	all	all
_time_allowed	Time allowed (seconds)	all	—

Item-level variables use the `corr` value-label family in the DDI codebook (typically 0 = Incorrect, 1 = Correct, with non-response codes).

Summary metrics (scores, rates, times) are usually continuous and unlabelled.

All variable-level codes and labels are documented in the portal **DDI codebook**.

8. References

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